Ordinary binary S3: actual absolute irreducibility

Result. The literal ordinary polynomial $(xy+xz+yz)^2+xyz+b$ is symmetric and irreducible over every characteristic-two field, and remains irreducible after every field scalar extension, for every b. Its evaluation is exactly the existing ordinary S3formula; on every nonsingular ordinary model $y^2+xy=x^3+a_2x^2+b$ over an algebraically closed field, vanishing is equivalent to an actual Mathlib three-point zero-sum. Signed standing unverified. [pcc:unverified]

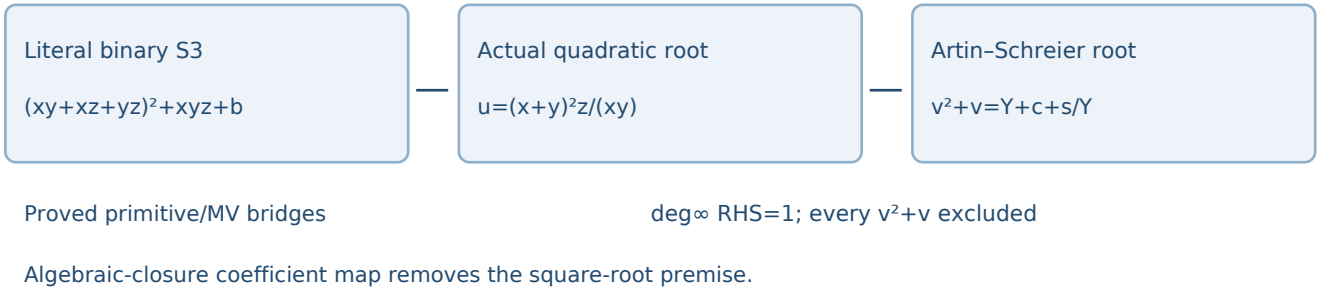

Figure. Actual polynomial and rational-function maps are proved identities. After adjoining $s^2=b$, a root would yield $v^2+v=Y+c+s/Y$, whose actual rational-function degree is1. For positive degree v, the left side has even degree2deg(v); for nonpositive degree v it has degree at most0. Both contradict1. No Artin–Schreier obstruction or irreducibility is assumed. [pcc:unverified]

Literal and scalar bridges. The actual polynomial is primitive: its leading coefficient is the diagonal square, while the middle coefficient xy is not divisible by that irreducible diagonal. The proved injective coefficient map into the algebraic-closure rational-function field and Gauss bridge descend irreducibility. Explicit MV/nested algebra equivalences, exact evaluation and scalar-map identities bind the result to the source polynomial, including all field extensions. The final theorem has no square-root or finite-enumeration premise. [pcc:unverified]

Negative control and skepticism. The distinct short-Weierstrass expression at F2,a=b=0 is a nonunit square and directly not irreducible. Lean proves it is unequal to the ordinary binary polynomial by evaluation. The control distinguishes models; it does not refute the qualified ordinary source. Algebraic irreducibility allows b=0, while the actual elliptic-group equivalence explicitly retains nonsingularity and ordinary model conditions, including repeated-coordinate/two-torsion cases. [pcc:unverified]

Checks. Whole gate43/0failures/0trusted. Lean4.35.0/Mathlib56a00585…. Complete closed actual_binary_S3_certificate, custom import graph, configs and declaration hash are pinned in sources.json; proof-map.json identifies the intermediate bridges. Failed/successful source/log/exit records are retained losslessly. Only propext/Classical.choice/Quot.sound; no holes, custom axioms, native evaluator, raised limits, global elaboration flags or waivers. Exact PCC receipt is separate; signed standing unverified. [pcc:unverified]

Sources. Semaev2015section2/arXiv1504.01175v1, actual PDFpage3 formula visually inspected. Kosters–Yeo2015section2general b-invariant formula/Proposition2.1 and4.2ordinary hypotheses. Original Semaev2004Theorem1is characteristic not2/3. Existing actual BinaryS3group equivalence and short-model square control, pinned Mathlib degree/algebraic closure/Gauss/MV APIs reviewed. KB unavailable, bounded primary/repository/library fallback; no novelty claim. [pcc:unverified]

Remaining scope. General higher-arity absolute irreducibility and recursive splits k≥3remain open. This packet confirms the ordinary three-variable formula and does not claim those broader theorems. No native experiment, paid compute or agent upload. All fourteen obligations remain in the current requirement map. [pcc:unverified]